Remote Sensing Image Super-resolution: Challenges and Approaches

(Invited Paper)

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***Abstract*—Remote sensing has a growing relevance in the modern society with the development of image processing of satellite imagery. However, due to the limitations of the current imaging sensors and the complex atmospheric conditions, we are facing great challenges in the remote sensing applications due to the limited spatial, spectral, radiometric and temporal resolutions. Therefore, super-resolution techniques have attracted much attention by which the low quality low resolution remote sensing images are enhanced. In this paper, we discuss the challenges in remote sensing image super-resolution and thereafter review the relevant approaches. More specifically, the different categories of remote sensing techniques, i.e., the learning-based, interpolation based, frequency domain based, and probability based methods, are reviewed and discussed. Furthermore, the super-resolution applications are discussed and insightful comments on future research directions are provided.**

***Index Terms*—Super resolution; remote sensing; observation model.**

I. Introduction

In remote sensing imaging applications, high-quality imageries are required and high resolution (HR) images are usually desired for remote sensing image analysis and processing procedure. However, RS images always distort due to the limitations of remote sensing image sensors and other factors like optical system aberration, atmospheric disturbance, movement, noise of imaging system *etc.* Meanwhile, the acquired low resolution RS images suffer from difficulties in many remote sensing applications such as map updating, road extraction and military target identification.

Therefore, it is very useful to construct HR images from low resolution (LR) images. So that the resolution of LR image with complementary information can be increased and the quality of images can be enhanced. The SR technology can break through the resolution limit of image acquisition equipment and realize the image reconstruction at sub-pixel level with complementary information of LR image sequences. Therefore, the SR technology can help in resource development, environmental monitoring, disaster research, and global change analysis.

The Super Resolution (SR) technology reconstructs an HR image with a single low resolution (LR) image or a sequence of LR images. Sub-pixel shifts of LR images sequences are often used in the SR image reconstruction. They can be acquired from the same area images of different sensors in the same time. The SR reconstruction technology was introduced by Harris [1] and Goodman [2] in 1960s, aiming to use a single LR image to reconstruct the HR images. The reconstruction idea of multiframe LR images was first proposed by Tsai and Huang [3] in 1984 in order to improve the spatial resolution of Landsat TM images. After years of development, the super resolution reconstruction technology has become mature and applied in many fields. The SR technology can realize good reconstruction with video images, static images, for their easy and simple registration model. But for remote sensing images, the registration is more difficult because of the complex topographic relief. The super mode in the SPOT-5 satellite system is one of the most successful applications. It overcomes the registration by using both the hardware method and the imaging processing approach [4].

The rest of the paper is organized as follows. In Section II, we firstly introduce the observation model end describe the degradation procedure of remote sensing images. Then four categories of SR image reconstruction technologies are discussed in Section III. Section IV presents the experiment results of different state-of-art SR image reconstruction methods. Section V concludes the paper, and section VI acknowledges the provision of the image data.

II. Observation Model

RS images suffer from various kinds of degradations during the process of acquisition. The degradation procedure can be modeled by four operations [4], [5], [6]. (i) Warping: translation, rotation and scaling. (ii) Blurring: a low-pass filter. (iii) Down-sampling by a factor of $q_1 \times q_2$. (iv) Noise: a great probability of white Gaussian. The model can be expressed in the form of matrix

$$y_k = D_k B_k W_k x + n_k,\ k = 1, 2, \ldots, z \tag{1}$$

where x is the original HR image with the size of $M_1 \times M_2$, z is the total number of LR images, W_k is the warping matrix which keeps the motion information during the image acquisition process, B_k is the blurring matrix which represents the blur factor caused by optical system problems, D_k is the down-sampling operator, n_k usually presents as Gaussian noise, and y_k is the kth LR image with a size of $L_1 \times L_2$ where $L_1 = M_1/q_1$ and $L_2 = M_2/q_2$.

Both of the HR image and LR images are represented in the lexicographic-ordered vector form with the size of $M_1 M_2 \times 1$ and $L_1 L_2 \times 1$, respectively. According to the Equation 1, the

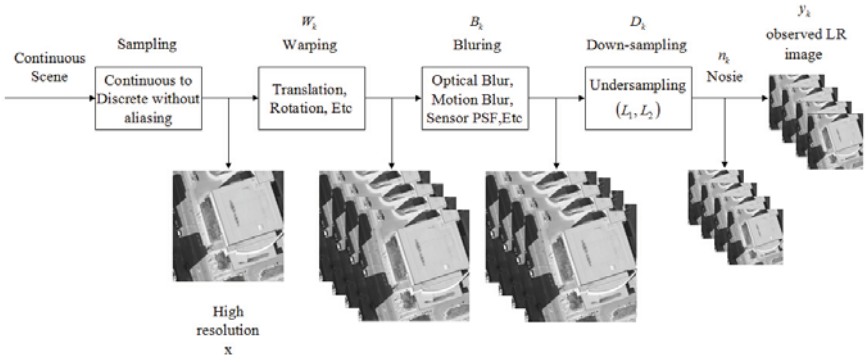

Fig. 1. Observation model relating HR image to LR images.

degradation procedure can simplified the model by making $A_k = D_k B_k W_k$. Then the formula is represented as

$$y_k = A_k x + n_k, \ k = 1, 2, \ldots, z \tag{2}$$

The observation model relating HR image to LR images is shown in Figure 1.

The SR reconstruction is a feasible reverse process of the image degradation. Hence, the solution is equal to the determination the matrix A_k. That is to determine: (i) Motion estimation to get W_k. (ii) Blurring estimation to obtain B_k. (iii) Noise estimation to determine n_k. All of the three matrices are sparse and should be estimated from LR images, the SR reconstruction is thus an ill-posed problem [7].

III. SR Models in Remote Sensing

Super resolution reconstruction methods are techniques which generate a high resolution (HR) image from a sequence of low resolution (LR) images. As mentioned earlier in this paper, the SR image reconstruction technology used in remote sensing can be divided into four categories: (i) the learning-based method; (ii) the interpolation method; (iii) the frequency domain based method ; (iv) the probability theory based method.

A. Learning-based SR model

The learning-based method is an important research direction of SR reconstruction nowadays. It aims to learn a prior knowledge of mapping from patches between LR and HR images in a training database. The mapping model can be a group of learned interpolation kernels, a lookup table of LR patches or a mapping coefficient between LR patches and HR patches.

The sparse representation method is one of learning-based SR methods. It was first proposed by Yang *et al.* in [8] and can be divided into three steps. Firstly, learning an over-complete dictionary from training patches simply randomly sampled in training set. Then, each test patch is expressed in the over-complete dictionary with sparse coefficient. Finally, reconstructing the HR image with the weighted coefficient. In [9], the sparse representation method is applied to reconstruct remote sensing images. The over-complete dictionary can be divided into two parts: The primitive dictionary pair and the residual dictionary pair. The first one is used to obtain the initial HR remote sensing image from a single LR image. The second one is a learning-based method which reconstructs the lossy information from the original HR image. It has been testified that, with this method, the resolution is improved efficiently.

The support vector regression (SVR) is a kernel regression method with a good ability of nonlinear mapping. Zhang *et al.* used a support vector regression (SVR) method to learn the prior knowledge between HR images and LR images in [10]. The interpolated images which lack of high-frequencies in [10] were over sampled to the same resolution and interpolated by using bilinear interpolation method. The prior knowledge learned from SVR was used to estimate the high frequency components when the HR image was reconstructed. The aim of [10] was to convert the LR ETM+ images into a single HR ASTER image. Experimental results of [10] have shown that even with a small amount of samples, SVR can also learn a model well.

B. Interpolation based SR model

Compared with the learning-based method, the interpolation method for the SR image reconstruction is a more intuitive one. It can be divided into three steps. (i) Registration step: basis of SR reconstruction. (ii) Reconstruction step: Interpolating the LR images into a regular HR grid. The interpolation method can be biquadratic interpolation, projection onto convex sets (POCS), Iteration Back Projection (IBP) and so on. (iii) Deblurring step: removing the blur and noise to improve the resolution of the reconstruction image. Among these three steps, step (ii) is the core step of interpolation SR. Step (i) and (iii) are auxiliary steps.

Aguena *et al.* [11] adopted the POCS method to do the interpolation in a fusion process. The POCS algorithm was first suggested by Stark and Oskoui [7] who successfully combined the prior knowledge into the model as convex sets. The goal of [11] was to get the combination characteristics of both multi-spectral (bands from CBERS-1) and panchromatic (bands from Landsat-7) images. The process can be divided into two steps: interpolation reconstruction and synthesis.

Here, we introduce the reconstruction process only. The most important feature of POCS is that the prior information of the solution can be easily incorporated into the model to find a point which satisfies the restrictions. The convex constraint set is composed by all the restrictions. By being iteratively projected onto convex set, the desired point can be found in the intersection which can be incorporated in the SR reconstruction image.

The iteration back projection (IBP) is also one of the most popular interpolation methods. The SR image is estimated by back projecting the difference between simulated LR images via imaging blur and the observed LR images in [12]. The reconstruction process is realized by minimizing the energy of the error iteratively. Based on the original method, Li *et al*. [13] applied a modified IBP on the remote sensing images and selected blur kernels separately for each sensor. Furthermore, different coefficients for the error images were adopted in the process of back-projection. SR reconstruction results would be obtained by experiments on a set of images generated from one ETM+ channel and a sequence of ALOS images.

C. Frequency domain SR model

The SR model in frequency domain solves the reconstruction problem by Fourier transformation and inverse Fourier transformation (or other transformations, like wavelet transformation). The main idea of the SR reconstruction in frequency domain is to enhance the high-resolution information intuitively with low computation complexity. It can be divided into four steps: (i) Image registration; (ii)Transformation into frequency domain; (iii) Reconstruction; (iv) Transformation back to spatial domain.

To improve the resolution of images in the frequency domain, we can increase the high-frequency component of images. One feasible way is to avoid the aliasing effect. Chen *et al*. [14] used a method of spectral de-aliasing to reconstruct a HR image. LR images are first generated by sub-sampling from the original HR remote sensing image and the fourier transformation. To get the translation parameters, the phase correlation method is adopted. With translation parameters and the spectral de-aliasing method, images can be reconstructed. Finally, the HR image is generated by inverse Fourier transformation. Compared with the bilinear interpolation method, the result of de-aliasing method performs better.

The wavelet-interpolation based method which combines the wavelet transform and interpolation was first proposed by Nguyen and Milanfar [15] [16]. In [17], Tao *et al*. first decomposed the remote sensing image with the DWT method. The interpolated wavelet coefficient images can be generated by nearest interpolation, bilinear interpolation or bicubic interpolation. The SR image is obtained by the inverse discrete wavelet transform. Results show that the wavelet-interpolation combination algorithm protected the high frequency information of the original HR image effectively.

The fourier-wavelet regularized deconvolution (ForWarD) is another method to recover an HR image [18] [19]. It uses scalar shrinkage in both fourier and wavelet domains to achieve the noise regularization and the reconstruction. The ForWarD is a two-step algorithm. The first step is the Fourier-based Weiner filtering which is to build a sharp images. The second step is to denoise by applying the stationary wavelet transformation. Several satellite images from NASA are tested. Results show that this algorithm can be applied to any resolution images to achieve the high resolution.

D. Probability theory SR model

With the fact that the SR reconstruction is an ill-posed problem, some priori conditions and restrictions must be imposed additionally to transform the SR reconstruction problem into a well-posed problem. Motivated by this, several methods based on the probability theory have been proposed. Bayesian method is widely used in the SR reconstruction because it can incorporate the prior probability density function with prior constraints. The basic idea is to take account of both LR observed images and the priori knowledge of unknown HR images, and get HR images with bayesian theory.

The maximum likelihood (ML) method is one of the most popular Bayesian-based methods, which was first proposed by Tom and Katsaggelos [20]. The solution is to find the ML estimation of HR images. The key point is to solve the probability density function (PDF). The maximum a posterior (MAP) method is another popular method based on Bayesian, which assumes that HR images can be acquired by maximizing its posterior probability under the condition of LR image sequences existence. Wang *et al*. [21] proposed a MAP method based on iterative optimization to reconstruct the SR images while keeping the spectral information of multi-spectral images. Apart from a set of SPOT XS LR images, a panchromatic image is also needed. With a high-pass filter, the high frequency information can be distilled which is used to combine with the interpolated multi-spectral images. After the calculation of mean value and variance based on mapping, a primary HR image will be obtained. Then the MAP based iterative optimization is proposed to receive a further improvement of resolution and avoid the quality decrease. With this method, the spatial resolution of the reconstructed HR image is enhanced with little spectral information loss.

The total variation (TV) is an effective SR method based on regularization with the ability of preserving edges. However, pseudo edges can also be preserved and produced smooth regions. With this shortcoming, Yuan *et al*. [22] proposed a regional spatially adaptive total variation (RSATV) model based on remote sensing images, which improved the pseudo edges obviously. Firstly, spatial information is extracted pixel by pixel and a median filter is used to enhance it. Spatial weight is then constructed and is filtered. After that, classification with K-means clustering is applied. With these two filtering processes, the pseudo edges is suppressed respectively. Finally, for each region, the clustering center value represents the regularization strength. So the total variation regularization is transformed from pixel-based to region-based. Tested on the

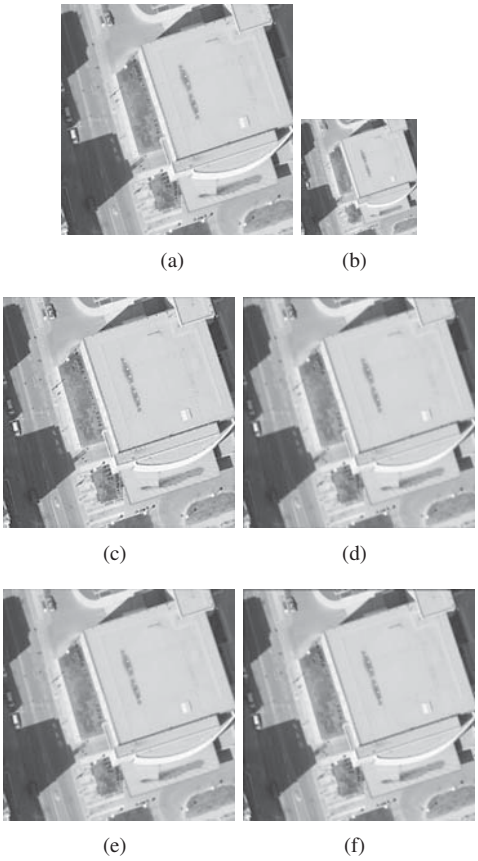

Fig. 2. Experimental results. (a) original HR image. (b) the first LR image. (c) Sparse Representation. (d) POCS. (e) Wavelet-Interpolation. (f) Tikhonov Regularization.

remote sensing, this method can not only preserve the edges but also is robust to the noise.

The tikhonov regularization is also a SR reconstruction method, which is named after Tikhonov A N [23]. It is a common way to resolve the ill-posed problem. With the method of adding, Tikhonov regularization generates a prior probability function of the original image on the base of the mean squared error cost function. Compared with other regularization methods, Tikhonov regularization can maintain the edges and smooth noise simultaneously.

IV. Experimental Results

In this paper, we use digital aerial images captured over Downtown Toronto downloaded from ISPRS to do remote sensing SR reconstruction experiments. As shown in Fig 2, the cropped image with the size of 512×512 is used as the original HR image to generate LR images. Based on the observation model shown in Fig 1, four LR images (each with size of 256×256) are obtained by shifting with subpixel precision and down-sampling with a zoom factor.

We adopt four methods to compare the effect of the SR reconstruction: Sparse Representation(SRP), POCS, Wavelet-Interpolation(WI), Tikhonov Regularization(TR). POCS and Tikhonov Regularization methods need a sequence of LR images to utilize the complementary information. We take all the four LR images to be the inputs. However, Wavelet-Interpolation and Sparse Representation methods only need a single LR image. So we take the first LR image as the input. Particularly, for the method of Sparse Representation, we use the official code of the paper [8]. The training data are 100 digital aerial images. We choose 50000 patches to train another dictionary. Results are shown respectively in Fig 2.

TABLE I
PSNR, MSE, SSIM AND TIME COST OF THE FOUR METHODS

	SRP	POCS	WI	TR
PSNR	35.83	34.22	36.80	33.03
MSE	16.99	24.62	13.59	32.37
SSIM	0.96	0.88	0.97	0.77
TIME(s)	43.28	2.61	1.45	7.92

To evaluate the quality of the reconstruction results, comparing with the original HR image, we adopt three quality evaluation factors: PSNR (Peak Signal to Noise Ratio), MSE (Mean Squared Error), SSIM (Structural Similarity). We also compute the time cost of each method. All of the evaluation indices are presented in Table I. Time cost is obtained where the CPU is Intel(R) Xeon(R) E5-2630 v2, a dual-core processor with 2.60GHz, with 64G memory.

According to the results in Table I, we can conclude that the WI method outperforms others on our test RS images. Moreover, the WI method is faster than other methods. It is noted that the training process of the SRP method takes long time to get the patch data and the dictionary where RS images were used for training.

V. Conclusion

In this paper, we present a comprehensive review of the super resolution reconstruction technology for remote sensing images. Four SR reconstruction categories are specifically introduced: the learning-based SR, the interpolation based SR, the frequency domain based SR, and the probability theory based SR. We also conduct experiments based on several state-of-art methods to present the SR reconstruction results for remote sensing images.

Because the SR reconstruction technology can overcome the inherent resolution limitation of the imaging system, it can effectively enhance the resolution of aerial images or satellite images. Thus, the SR reconstruction technology has been proved practicability and important value in remote sensing field and it is widely used in identification and localization, image fusion, environmental monitoring, disaster research and global change analysis, *etc.*

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